

Saadhika Dhaniyala

Potsdam, NY | saadhikadhaniyala@gmail.com | (315) 244-8939

Education

Potsdam High School

Expected June 2026

The Clarkson School — Early College Program (GPA: 4.0)

- **GPA:** 98.80 (Unweighted), 100.53 (Weighted)
- **ACT:** 35 (E 35, M 35, R 36, S 34)
- **Advanced Coursework:** AP Biology (5), AP Chemistry (5), Genetics, Biostatistics, General Chemistry I & II, Physics I & II, Calculus I & II, Introduction to Programming

Honors and Awards

- **Best Poster & Presentation — Clarkson RAPS Conference** (August 2025) — Selected from top undergraduate students for clarity of scientific communication, experimental design, and research significance
- **Science Olympiad Medalist** (2022 – 2025) — Earned 11 medals across biology, earth science, and engineering events at regional and state competitions
- **Scholar Athlete** (2022 – Present) — Recognized for academic excellence while competing in varsity soccer, swimming, and indoor/outdoor track
- **National Rural and Small Town Recognition Program** (2024, 2025)
- **AP Scholar with Distinction** (2025)
- **National French Honor Society** (2024 – Present)
- **National Art Honor Society** (2022 – Present)

Research Experience

Research Assistant — Clarkson University, Dept. of Chemistry & Biochemistry

June 2025 – Present

- Developed **biodegradable, surface-eroding bone cement formulations** using cyclic poly(methacrylic anhydride) (cPMAA) and PEGDMA for controlled drug delivery
- Synthesized polymer networks via **redox initiation (BPO/DMT)** and cured crosslinked cement matrices
- Evaluated degradation under **physiological conditions** (pH 7.4, 37°C), observing surface erosion with preserved geometry
- Quantified **erosion-driven drug release kinetics** using UV-Vis spectroscopy with purpurin as a hydrophobic model compound
- Demonstrated **tunable degradation rates** through hydrophilicity and crosslink density modulation

Leadership & Activities

- **Medical Club Founder & President** (2023 – Present): Founded schoolwide organization; raised \$3,700 to fund cadaver lab visits, clinical workshops, and guest speakers
- **Class of 2026 President** (2023 – 2025): organized Fall Clean-Up initiative, Krispy Kreme, bakery, bottle drive, talent show, and school dance fundraisers
- **National Honor Society President** (2024 – Present): Led peer tutoring initiatives and coordinated community service
- **Science Olympiad Co-Captain** (2022 – Present): Mentored members across biology, earth science, and engineering events
- **Student Council Representative** (2024 – Present): Assisted with school improvement initiatives
- **Clinical Shadowing — Family Medicine & Pulmonology** (2025 – Present): Observed diagnostic reasoning and patient communication across rural primary care and specialized respiratory medicine. Witnessed how physicians manage chronic conditions and the physical and emotional impact of illness on patient identity

Athletics & Arts

- **Figure Skating:** USFS Gold Medalist (Dance); Senior-level skater training International Dances; 20+ gold medals at Empire State Winter Games (Largest winter sporting event in the Northeast); Qualified for State Games of America (2024, 2026)
- **Varsity Indoor & Outdoor Track:** Sectional qualifier (Triple Jump)
- **Varsity Soccer** (2024–Present)
- **Concert Band:** 1st Chair Alto Saxophone; NYSSMA Soloist; Area-All County qualifier (2019–Present)

Work & Service

- **State Certified Lifeguard — Higley Flow State Park & Clarkson University Pool** (2023 – Present): 40+ hrs/week during the summer; responsible for swimmer safety, emergency response, and enforcement of aquatic safety protocols
- **Peer Tutor — LearnToBe & NHS:** Online and in-person tutoring for low-income students (2022–Present)
- **North Country Children's Museum:** Volunteer (2022–2023); STEM Education Intern (2023–2024)
- **Learn-to-Skate Volunteer Coach:** Assisted beginner skaters twice weekly

Skills

- **Laboratory:** Polymer synthesis, redox polymerization, spectroscopy, degradation studies, scientific poster presentation
- **Data:** Experimental design, data interpretation, graphing & visualization
- **Tools:** LaTeX, Microsoft Excel, PowerPoint, Google Workspace
- **Leadership:** Fundraising, mentoring, public speaking, event logistics